

Lab Experiment: 02

Lab Assignment 1: Understanding Union vs Structure

```
#include <stdio.h>
```

```
#include <string.h>
```

```
struct StudentStruct {  
    char name[20];  
    int studID;  
};
```

```
union StudentUnion {  
    char name[20];  
    int studID;  
};
```

```
int main() {  
    struct StudentStruct stuStruct;  
    strcpy(stuStruct.name, "Krishna");  
    stuStruct.studID = 0011;  
  
    union StudentUnion stUnion;  
    strcpy(stUnion.name, "Krishna");  
    stUnion.studID = 0011;  
  
    //displaying sorted values in structure  
    printf("Using structure:\n");  
    printf("Name: %s\n", stuStruct.name);
```

```

printf("Student ID: %d\n", stuStruct.studID);

// Displaying stored values in union
printf("\nUsing union:\n");
printf("Name: %s\n", stUnion.name);
printf("Student ID: %d\n", stUnion.studID);

// Display memory size occupied by structure and union
printf("\nSize of structure: %lu bytes\n", sizeof(stuStruct));
printf("Size of union: %lu bytes\n", sizeof(stUnion));
}

```

```

Using structure:
Name: Krishna
Student ID: 9

Using union:
Name:
Student ID: 9

Size of structure: 24 bytes
Size of union: 20 bytes

=== Code Execution Successful ===

```

Lab Assignment 2: Dynamic Memory Allocation with malloc() and free()

```

#include <stdio.h>

#include <stdlib.h>

int main() {
    int n, *arr, sum=0;
    float avg;

```

```
// Input the number of elements
printf("Enter the number of elements: ");
scanf("%d", &n);

// Dynamically allocate memory using malloc()
arr = (int *)malloc(n * sizeof(int));

// Check if memory allocation was successful
if (arr == NULL) {
    printf("Memory allocation failed!\n");
    return 1; // Exit if memory allocation failed
}

// Input elements into the array
printf("Enter %d elements:\n", n);
for (int i = 0; i < n; i++) {
    scanf("%d", &arr[i]);
}

// Calculating the sum of the elements
for (int i = 0; i < n; i++) {
    sum += arr[i];
}

// Calculate the average
avg = sum / n;

// Output the sum and average
```

```
printf("Sum: %d\n", sum);  
printf("Average: %.2f\n", avg);  
  
// Release the dynamically allocated memory  
free(arr);  
  
return 0;  
}
```

```
Enter the number of elements: 3  
Enter 3 elements:  
11 3 55  
Sum: 69  
Average: 23.00  
  
=== Code Execution Successful ===
```